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EviChain: Blockchain-Based Evidence Assurance

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Abstract

The management and preservation of digital evidence is an urgent matter and it is among one of the most important contemporary problems in the sphere of forensic investigations. Centralized databases are still in use by the law enforcement agencies in Pakistan and other parts of the world to store evidence obtained at the scenes of crime, during cyberattacks and financial fraud. The biggest weakness of these systems is that they can be tampered with and a sure and certain way of proving that the evidence was in its original form when it was originally collected does not always exist.

EviChain is a project whereby the authors seek to answer this question using two partially more refined technologies of the past couple of years blockchain, and the InterPlanetary File System or IPFS. The idea is one straight forward. In a system, it is believed that after an investigator uploads a body of digital evidence, the file is sent using a 64-character(s)-long fingerprint (SHA-256 hash) generating algorithm that generates a fingerprint that is unique to that file. Although one or two bytes in the file may be modified later, this fingerprint will not be similar in any way and the manipulation will be instantly noticed.

This is because the file is then stored on IPFS as opposed to being stored on one server. IPFS distributes files to a distributed network of nodes and uses each file to store a Content Identifier, or CID, based on its content, as opposed to its physical location. The CID and file hash, upload timestamp and the identity of the individual submitting the evidence are then stored into an Ethereum smart contract. Smart contracts are programs within the blockchain which cannot ever be modified once they are deployed. This implies that the trace of the evidence is permanently embedded in a community, verifiable log.

The outcome is a system wherein evidence integrity is not merely espoused within an institution but a system that could be mathematically demonstrated by any party including defence counsel, judge and freestanding forensic experts. The smart contracts of EviChain were written in Solidity, the backend in Python and Flask, and the frontend interface in React. The test was conducted on Ethereum Sepolia testnet.

CHAPTER 1: INTRODUCTION

1.1 Background

Any person, who has trailed a court case that has dealt with digital evidence, would likely have heard claims on whether the evidence was modified before it could be presented. Theoretical concerns are not the only concerns. Incidents in the world where the digital evidence stored in police or forensic laboratories were discovered to have been altered either due to ignorance or malabsorption have been recorded. The results go as far as an innocent person is convicted, or a truly guilty person acquitted.

The main reason of these failures is not hard to be found. Digital evidence is currently primarily stored in the manner in which regular files are stored: Even on servers that are operated by an individual organization. Whomever possesses the server owns the evidence. Evidence can be altered or erased by a corrupt official, a hacker or even a mere software bug and that will not leave any noticeable evidence that would have been noticeable to an external observer. Chain-of-custody, the record of who touched what and when, is usually stored as a spreadsheet or database record which is equally easily altered, as is the evidence itself. The underlying context of Blockchain technology, cryptocurrency, was developed in an entirely different context, although the same properties that make it great to record financial transactions are the same that make it incredibly useful in recording evidence metadata. When it has been written to a blockchain such as Ethereum that is public, nobody can alter it. All the network nodes have a copy of the ledger and any effort made to change any record would be automatically discredited by the consensus mechanism.

IPFS provides a new aspect to this image. Conventional file storage locates files based on their location and this implies that a file can be replaced by another file at the same location without any informing message. IPFS recognizes files based on their content. The addresses of files stored in IPFS are literally the hash of the contents of the files using cryptography. When the file alters then its address alters as well. This renders any interference to files of stored evidence instantly and without any bias to detect.

These two technologies are combined by EviChain, and an application interface designed for application by forensic experts is created. It is intended to be employed by individuals with evidence knowledge, yet who are not necessarily highly technical experts in blockchain or distributed systems. This is because an investigator should be in a position to upload a file, get a confirmation that it has

been conserved and subsequently confirm that it has not been modified without having to understand the security cryptographic processes involved.

1.2 Problem Statement and Objectives

1.2.1 Problem Statement

Current digital evidence management systems save evidence on central servers where it may be altered, removed or inherited without any professionally language verifiable mark of the character of the occurrence. The documentation on chain-of-custody exists in mutable forms and can be manipulated. No generally accepted system is in place that gives a mathematically assured, independently verifiable history of evidence integrity since the time of collection and beyond.

When such a mechanism is not present, it poses significant weaknesses in the justice process. The insiders can tamper with the evidence. Servers are prone to failure and loss of irreparable information. Defence teams also cannot have an objective way of making sure that evidence that was provided in the court is the evidence that was initially seized. They are not hypothetical risks or failures but are documented instances which have derailed judicial process in various jurisdictions.

1.2.2 Objectives

This project set out to accomplish the following:

- To build a working digital evidence preservation system that uses blockchain and IPFS to eliminate the vulnerabilities of centralized storage.
- To implement SHA-256 hashing so that every uploaded file receives a unique, verifiable fingerprint at the moment of upload.
- To store evidence files on IPFS so that they are distributed across multiple nodes and cannot be lost or altered at any single point.
- To record evidence metadata on an Ethereum smart contract so that the record becomes permanently immutable and publicly verifiable.
- To develop a web interface that allows forensic investigators to upload, view, and verify evidence without requiring specialist blockchain knowledge.
- To provide automated chain-of-custody tracking that creates a tamper-proof log of every interaction with a piece of evidence.

- To test the system thoroughly, including security and performance evaluation, and to document findings in a formal report.

1.3 Why This Project Matters

This project is of importance at the cross-section of justice and technology. Technically speaking, it exemplifies a practical implementation of two distributed computing concepts: blockchain and peer-to-peer file systems. Legally and forensically, it fills a real gap in the evidence management infrastructure utilized by law enforcement and the courts.

The area of the digital forensics in Pakistan, in particular, is still emerging. Cases involving digital evidence are becoming more common in courts, in both cybercrime and financial fraud and even when it comes to investigating terrorism, but the infrastructure to handle such evidence has lagged behind. Even in its present early state, a system such as EviChain is an indication of what forensic evidence management might be like once it is built with mathematical integrity assurances, instead of institutional trust as its base.

This project is directly academically relevant as well. It builds upon and encompasses cryptography, distributed systems, smart contract programming, web development, as well as forensic science, some of the fundamental knowledge areas of the BS Digital Forensics and Cyber Security programme.

1.4 Scope

The project at the given stage of development encompasses the design of the main components of the system and their initial implementation. These consist of the SHA-256 hashing snippet, IPFS support to store files in a decentralized manner, Ethereum Smart Contract writing and execution in the Sepolia testnet, and the simple web interface of investigators. The system is intended to support forensic investigators and law enforcement agents who are involved in the investigation of crime and work with digital evidence.

The features added to the system, which are currently not present at this level, are integration with existing national case management or laboratory information system, support of other types of evidence than common file formats, multi-jurisdiction access control, or complete mobile compatibility. These are yet to be developed areas. The system also works on a testnet instead of

Ethereum mainnet, which inhibits real-world deployment but is suitable in the situation of the academic evaluation.

1.5 Report Structure

The current level of development of this report is separated into three chapters. This chapter, Chapter 1, has presented the project, outlined the problem being tackled in the project, outlined the objectives and the scope of the project. Chapter 2 discusses the issue in more technical and legal depth, paying close attention to what areas the current evidence management systems fail on and what are the implications of such failures. Chapter 3 surveys published literature on the topic of the project, such as studies on blockchain usage in forensics, the IPFS protocol, the digital evidence standard, and similar solutions, and determines the particular gaps that the EviChain project is meant to address. The complete software requirements specification, detailed system design and architecture, implementation details and code, test approach and testing protocols, as well as conclusions and future plan of the project will be presented in subsequent chapters, which will be done in the final submission.

CHAPTER 2: PROBLEM DEFINITION

2.1 Overview

Chapter 1 highlighted the overall issue of evidence integrity in digital forensics. This chapter focuses on that issue much more extensively. This is meant to furnish an accurate description of what is amiss in the existing evidence management practice, why those defects are hard to remedy with the present centralized architecture, or what are the specific implications of these failures. To assess the correctness of the solution presented by EviChain on the question of whether the solution has the right issues or not, it is only possible with knowledge of the problem.

2.2 How Current Evidence Management Systems Work

When such digital evidence is obtained (be it on the computer of a criminal, via a system, or in one of the logs of a network, or the location of a crime), it is typically stored into a purpose-built storage system also under the control of the investigative body. It may be a particular forensic lab server or as simple as a shared network drive or even the police department internal database. The evidence files are registered in a database and there is a record of a chain of custody form which is signed usually by a paper or an electronic trace in a case handling systems indicating the name of the individual collecting evidence, the time and source of that evidence.

The issue is that all the elements of this system can be changed. The files can be overwritten or deleted. The records which are held in the database can be modified. Chain of custody has the potential to be falsified or lost. And above all, there is no system, as a matter of course, by which an outsider can determine whether any or all of these things had taken place, except, again, a forensic inspection of storage system on its own merits, which has its own chain-of custody problems.2.3 Specific Vulnerabilities in Centralized Systems

2.3.1 Insider Threat

The greatest threat in centralized evidence management is the insider threat. Technically, any user who has administrative access to the storage service of evidence, be it an IT administrator, high tier investigator or even a contracted maintenance professional has the technical capacity to alter or destroy evidence. The presence of access control systems and audit logs are a sort of protection, but they too are under control of the same organization that controls evidence, which poses an apparent

conflict of interest. Even in highly recorded corruption by police, the police officers who are mandated to preserve evidence have been found to have planted or destroyed evidence.

2.3.2 Single Point of Failure

A single point of failure is between the centralized servers. The digital evidence may be permanently lost due to a hardware malfunction, a ransomware virus, a fire in a data centre, or a poorly set-up backup system. Data on a corrupted or encrypted server can end up as totally non-recoverable, unlike in the case of physical evidence, which can often survive fires, floods etc. A number of high-profile criminal cases have been undermined due to the loss of digital evidence in precisely such manner.

2.3.3 Absence of Cryptographic Integrity Verification

Most current evidence management systems do not compute cryptographic hashes on evidence files when collected and check them each time the files are accessed after that. Other systems generate a hash during the uploading process, but the hockey stick is retained in the same database along with the evidence file and the idea is that in case the evidence file is altered then the hockey stick can be altered accordingly (the stored hash can be altered). There is no objective means of demonstrating that there has been no alteration of the evidence unless there is a separate record of the original hash that is independently verifiable.

2.3.4 Mutable Chain-of-Custody Records

Chain-of-custody records are most commonly stored in the form of a written or electronic record that may be modified retrospectively. An individual who has access to the case management system can in principle add or remove or alter entries to the chain-of-custody log without any technical obstacle. Practically, this implies that chain-of-custody documents stand as credible as the institution, which keeps them, and it is the institution whose actions can be questioned.

2.4 Legal and Judicial Consequences

The above-mentioned technical vulnerabilities directly have implications in court. The admissibility of digital evidence in a court of law demands under most legal systems that the evidence must be proven to be authentic, and that the integrity of the evidence can be secured by proving through an intact line of custody. The courts are confronted with a number of options, which are not good, in view of the concept of justice, when these requirements cannot be met.

Evidence which is unable to demonstrate authenticity can simply be omitted and this may of course sink a prosecution of which the omitted evidence would have easily proved guilt beyond reasonable doubt. In other cases courts may allow evidence of dubious honesty but direct the juries to handle it with care, bringing uncertainty to the result. Challenges of digital evidence are increasingly succeeding by defence counsel, and the lack of cryptographic integrity verification is a desirable target, in part because it is a real technical vulnerability and not just a procedural one.

Most importantly, because of the lack of transparency in centralized evidence management platforms, a scenario in which false convictions might take place is created. When it is possible to plant or fiddle with evidence undetected, innocent individuals may be convicted based on the fabricated evidence. Criminal justice history is well-educated in cases of precisely this occurring that it cannot be poached as a purely theoretical issue.

2.5 Comparative Analysis of Existing Solutions

Vulnerability	How It Occurs	Consequence
Insider tampering	Admin modifies files without detection	Evidence inadmissible
Server failure	Hardware crash or ransomware destroys data	Permanent evidence loss
No hash verification	File changes go undetected	Tampered evidence admitted
Mutable audit logs	Chain-of-custody records edited	False documentation
Weak access controls	Unauthorised access to evidence	Evidence exposure or theft
No independent audit	Only controlling body can verify	No external accountability

2.6 What Existing Commercial Tools Offer and Where They Fall Short

Various commercial digital forensics systems are being used by law enforcement currently across the globe. The Cellebrite UFED, Magnet AXIOM, and FTK are all well-regarded tools with their abilities to obtain, process, and analyse the digital evidence on devices. These tools, however, were created in order to obtain and analyze evidence and not to store it and ensure its integrity over time. All the large commercial platforms use centralized storage of the evidence they process. None of them incorporate such blockchain or distributed file systems to preserve evidence. Others create cryptographic hashes on evidence files, a good practice in itself but hashes are maintained in the same system as the evidence thus their value as a free stand-alone integrity assurance. None of them

offers an independently verifiable chain-of-custody record that is not subject to tampering and thus cannot be audited by anyone not affiliated to the organization conducting the investigation.

The open-source software, such as Autopsy and Sleuth Kit, is also aimed at analysis, not preservation, and is also subject to the same architectural restrictions. The verdict after reviewing the tools at hand is that currently, no widely accepted answer exists that will give mathematically certain and independently verifiable evidence integrity between the point of collection till presentation in court.

2.7 Summary of the Problem

Summing up the strands of this chapter, the issue, which EviChain is aimed at, can be thus formulated. Electronic evidence kept in central databases can be modified by insiders, lost with a hardware malfunction, and tampered with without being detected at each contact point between seizure and presenting the evidence in court. Chain-of-custody trails are not verifiable but they can be changed. None of the currently available mainstream tools offer a secure cryptographically-backed, publicly viewable integrity history of digital evidence.

EviChain has suggested to mitigate through the use of three technologies: An unforgeable fingerprint of every evidence file by using SHA-256, Storing files in a distributed and inaccessible fashion (without any points of failure) using IPFS, and creating an unrestricted, publicly verifiable history of evidence metadata and chain-of-custody events using Ethereum smart contracts. The next chapter explores the scholarly and technical literature which underlies this approach and situates EviChain in the context of existing body of research.

CHAPTER 3: LITERATURE REVIEW

3.1 Purpose of This Review

The chapter is a survey of the academic and technical literature that is most pertinent to EviChain. It discusses four key topics, namely the history of blockchain technology and its potential use to address issues related to data integrity, the introduction of IPFS and content-addressed stores as an alternative to traditional file systems, the academic literature on the topic of managing digital evidence and a chain-of-custody, and an overview of systems suggested in the literature that can in

some sense be comparable to EviChain. The chapter ends by defining the exact gaps in the current research and practice, which is addressed by the given project.

The evidence used within the review is mostly peer-reviewed studies published in the last 2008 2023 timeframe and which can be supported by technical documentation in the form of the Bitcoin and Ethereum whitepapers and IPFS specification paper. It also appeals to applicable standards of forensic science in particular, the ISO/IEC 27037 on handling digital evidence.

3.2 Blockchain Technology and Immutable Record-Keeping

3.2.1 Origins and Core Properties

Nakamoto introduced the concept of the blockchain in 2008 as the foundation of the Bitcoin payments. The crucial idea was that a decentralized network of participants might keep a decent ledger containing no central authority, through a mixture of cryptographic connecting of records and an economic incentive mechanism to keep participants honest. The cryptographic hash of the last block is stored in each block of the chain, so the only reason one of the blocks of history can be changed is that every other block would have to be recomputed and a significant majority of the network would need to recognize the changed chain. Practically, on a large scale such as on a Bitcoin or Ethereum network, this is computationally infeasible.

It is precisely these properties, immutability, decentralization, and transparent verifiability that are interesting regarding the applications of blockchain technology far beyond the cryptocurrency industry. Any scenario that requires keeping of a permanent, mistrustful record of occasions, could be a good target of blockchain. An apparent example is evidence management.

3.2.2 Ethereum and Smart Contracts

In 2014, Buterin developed Ethereum as a blockchain technology that has an embedded programming language that can enable developers to execute what he termed as smart contracts. A smart contract is simply a program which runs on top of the blockchain itself and which is permanently stored on-chain and which automatically triggers when specific conditions are fulfilled. Since the code of contract and its output are pulled to the blockchain, they also exhibit the same as guaranteed by the blockchain to the ledger as a whole.

In the case of EviChain, the concept of smart contracts is used to permanently store evidence metadata. A smart contract can be created to receive an evidence hash, CID, time, and identity of the uploader, store it on the blockchain and render it forever readable and verifiable. After deploying

the contract and writing a record one cannot change the contract, even yourself who wrote a contract. This is an entirely new model of trust in comparison to whatever database model.

3.2.3 Blockchain in Forensics and Legal Applications

In 2019, Lone and Mir released what is perhaps the first significant academic proposal of blockchain-based forensic evidence management. In their article, Forensic-chain they suggested the implementation of Ethereum, which could serve as a chain-of-custody record of digital evidence. They showed that using smart contracts it was possible to record evidence hashes and access events such that they would independently be verifiable by any party with access to the Ethereum network. The technical feasibility of the approach which EviChain is based on has been proved by their work. In 2019, Brotsis and colleagues presented a study comparing the use of the private and public blockchain architectures in digital forensics within the Internet of Things settings. They have concluded that, transparency and verifiability of public blockchains is better than private ones since any individual can view the ledger without relying on the trust to the institution that controls the running of the ledger blockchain. This conclusion is a clear rationale behind EviChain architectural choice which is to develop on a publicly available blockchain as opposed to that which is privately available.

Rathore and co-authors investigated evidence verification systems based on smart contracts, and discovered that verification queries are under two seconds, a sufficient time to be applied in investigative processes. They were limited, however, to a similar restriction as that emerges across this literature, namely that the evidence files themselves were still stored in centralized databases, with only the metadata stored on-chain. This implies that the files within it are still susceptible to the same server-side risks mentioned in Chapter 2.

3.3 IPFS and Decentralized File Storage

3.3.1 The IPFS Protocol

Benet designed IPFS and explained it in a technical paper of 2014. The basic break with traditional file systems is the change in addressing by location to addressing by content. A file name in a traditional system is based on its location, e.g. a URL or file path. The file may be overwritten with an entirely new file without the system being aware of this. With IPFS, a file is given an address known as a Content Identifier, or CID, which is a cryptographic hash of the contents of the file. This

implies that there is no separation between the address and content. Any modification made in the file will generate another CID and the initial CID would not be able to reference the altered one. This is a very valuable property in the case of evidence management. A CID on the blockchain is not simply an address of a place where a file could be placed. It is a cryptographic evidence that the file of that address contains certain content. A researcher is able to access the file any time, and confirm that it contains the same content as what was saved to the blockchain, which presents an objective integrity verifiable that does not rely on the belief of any given institution.

3.3.2 IPFS in Practice

Ali and colleagues have shown IPFS application in a healthcare setting in storing sensitive records safely. The relevance of their work is due to the fact that healthcare records bear certain significant similarities with forensic evidence: they need to remain secret, they need to be proven to be untouched, and their access needs to be strictly regulated. Ali et al. reported that IPFS offered the required durability and integrity measures coupled with the practical implementation in a institutional context.

A supply chain provenance system by Nizamuddin et al. paired IPFS with Ethereum to store documents with IPFS and information about their CIDS with Ethereum. This design is structurally similar to what EviChain is instantiating and this fact offers helpful validation that the integration of the approach is not solely technically appealing, but also not just in theory appealing. Their system proved that files that have been stored at IPFS and managed by Ethereum smart contracts could be consistently accessed and checked in a variety of contexts of access.

3.4 Digital Evidence Management and Forensic Standards

3.4.1 Foundations of Digital Evidence Science

The thorough discussion of digital evidence and computer crime, which Casey presented in a number of editions most recently published in 2011, is a source of inspiration in the field. Casey sets out that digital evidence differs as compared to physical evidence in a crucial aspect; that it can be altered and traces not be left behind. One can edit a document, edit a log file, edit a photograph and no one ever gets a trace of these edits appearing on the file itself. That is why Casey contends that technical measures to ensure integrity, namely cryptographic hashing and highly constrained chain-of-custody are not optional additions but key prerequisites of evidence that can be scientifically and legally defensible.

In 2003, Carrier and Spafford presented an influential model of digital investigations, which discovered evidence preservation to be a crucial step that demanded certain technical protection. The line of thought in their model has led to the way the digital forensics community comprehends the evidence lifecycle between the gathering of evidence and its presentation in a court of law. EviChain can be interpreted as a real-world implementation of the preservation protection as discussed by Carrier and Spafford in theory.

3.4.2 Cryptographic Hashing in Forensic Practice

It is not a new development in forensics that hash functions are used to establish evidence integrity. Over a long period of time, MD5 and SHA-1 have been stalwart hashing MD5 and SHA-1 functions as the standard algorithms to verify evidence. They both were later shown to have collision weaknesses, that is, it was theoretically and practically possible to generate two distinct files based on the same hash value. This rendered them useless as integrity proofs. SHA-256 is now seen as the recommended standard for evidence hashing by NIST guidelines and international forensic practice, producing a 256-bit hash value and not known to have any practical collision attacks. SHA-256 is in use in EviChain.

3.4.3 International Standards

The digital evidence identification, collection, acquisition, and preservation guidelines that are internationally accepted are the ISO/IEC 27037:2012 standards. The criterion of standard term states that the integrity of the evidence has to be proven by the objective technical method and the chain-of-custody records must be kept in a manner that is reproducible and verifiable by other parties. These are exactly what EviChain is meant to provide. SHA-256 hash use also meets the integrity verification criterion and documenting evidence metadata on a public blockchain meets the independently verifiable chain-of-custody documentation requirement.

3.5 Comparative Review of Proposed Systems

Feature	Traditional System	Lone & Mir 2019	Nizamuddin et al. 2019	Rathore et al. 2021	EviChain
Blockchain metadata recording	No	Yes	Yes	Yes	Yes
IPFS file storage	No	No	Yes	No	Yes

Feature	Traditional System	Lone & Mir 2019	Nizamuddin et al. 2019	Rathore et al. 2021	EviChain
SHA-256 evidence hashing	Sometimes	Yes	No	Yes	Yes
Smart contract verification	No	Partial	Yes	Yes	Yes
Automated chain-of-custody	No	Partial	No	Partial	Yes
Web-based investigator interface	Varies	No	No	No	Yes
Publicly auditable records	No	Yes	Yes	Yes	Yes

3.6 Gaps Identified in the Literature

Examining the body of research that has been reviewed in this chapter, some gaps are identified as finding uniformity across the research literature.

The greatest shortcoming is lack of end to end decentralization. The table of proposals analyzed above demonstrates that the majority of suggested systems focus on either decentralized storage or metadata recording based on blockchains but not both. An example is Lone and Mir Forensic-chain where evidence hashes are recorded on Ethereum, but there is no mention of where evidence files reside. Rathore and co-authors also apply blockchain to metadata, storing the files as usual. This leads to a scenario whereby blockchain record will be able to identify whether there has been a modification in a file, but cannot intercept file deletion or loss of files altogether.

A second void is a lack of an operational user interface targeted at forensic investigators. The systems presented through the academic literature are commonly prototypes constructed by the researcher and need technological skills in operation. No reviewed papers describe a user interface that is based on a real world workflow of a forensic investigator submitting, retrieving, and verifying evidence within the context of a real world investigation.

Third, the literature discusses alignment with established standards of forensics very little. The majority of older suggested blockchain-based forensic systems are computer science in nature without a clear specification of what the legal and procedural constraints are that an evidence will actually be admissible in court. The ISO/IEC 27037 requirements are clearly considered in EviChain.

Lastly, the absence of practical implementation on running blockchain networks. The majority of the current literature employs either private or simulated blockchain settings. The implementation of EviChain on the Ethereum Sepolia public testnet, although not on the mainnet, more closely resembles a real environment as compared to most of the related work in the literature.

3.7 Summary

It is empirically validated in literature that the combination of blockchain technology and decentralized storage is a technically viable and more and more research-informed system to the issue of digital evidence integrity. EviChain incorporates the components of the solution presented by Research by Lone and Mir, Research by Brotsis and other researchers, Research by Nizamuddin and other researchers, Research by Rathore and other researchers. The evidential and the procedural requirements that any such technical solution must meet are given in the forensic science literature in the works of Casey and Carrier and Spafford and the ISO/IEC 27037 standard.

What the literature currently lacks is an integrated system that integrates IPFS-based decentralized file storage, Ethereum metadata recording of smart contract interfaces and SHA-256 evidence hashing, automated chain-of-custody tracking, and a specifically forensics-focused investigator interface, built directly around meeting international forensic evidence standards. EviChain, is an effort to bridge that gap.

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